ASSESSING AND FORECASTING GROUNDWATER RESOURCES IN THE CONTEXT OF CLIMATE CHANGE

PHASE I REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled "**ASSESSING AND FORECASTING GROUNDWATER RESOURCES IN THE CONTEXT OF CLIMATE CHANGE**" is the bonafide work of **HARIRAM S (2024207029)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT – ENGLISH

Tiruppur District, the "Knitwear Capital of India," faces a severe groundwater crisis driven by rapid industrialization, agricultural demand, and climate variability. Overextraction and contamination from textile processing have caused alarming declines in groundwater levels and quality, demanding an intelligent and adaptive forecasting solution.

This study develops a hybrid AI-based framework for assessing and forecasting groundwater resources under changing climatic conditions. The approach integrates multi-source datasets—including historical groundwater, meteorological, and IoT sensor data—and applies advanced signal decomposition techniques such as ICEEMDAN and VMD to manage non-stationary time-series behavior. Optimized LSTM networks, enhanced through the Slime Mould Algorithm (SMA), and CNN-LSTM hybrids are used to capture complex spatial-temporal relationships.

In Phase I, models are trained and validated on data from 1994–2024 using MAE, RMSE, R^2, and NSE metrics. Phase II integrates real-time monitoring and climate projections (CMIP6 SSP2-4.5 and SSP5-8.5) through an Adaptive Weighting Model, improving long-term predictive reliability.

The resulting system delivers accurate, interpretable groundwater forecasts, offering a data-driven decision-support tool for sustainable water management in Tiruppur and a scalable model for other climate-vulnerable regions.

ABSTRACT – TAMIL

இந்தியாவின் “நூலாடைத் தலைநகரம்” என அழைக்கப்படும் திருப்பூர் மாவட்டம், வேகமான தொழில்துறை வளர்ச்சி, வேளாண்மைத் தேவைகள் மற்றும் காலநிலை மாறுபாடுகள் காரணமாக கடுமையான நிலத்தடி நீர் நெருக்கடியை எதிர்கொள்கிறது. அதிகப்படியான நீர்எடுப்பு மற்றும் துணி தொழில்துறை மாசுபாடு, நிலத்தடி நீரின் அளவு மற்றும் தரத்தில் ஆபத்தான வீழ்ச்சியை ஏற்படுத்தியுள்ளது. இதற்கு புத்திசாலித்தனமான மற்றும் தன்னிச்சையாகச் செயல்படும் முன்னறிவிப்பு அமைப்பு அவசியமாகிறது.

இந்த ஆய்வில், காலநிலை மாற்றத்தின் சூழலில் நிலத்தடி நீர் வளங்களை மதிப்பீடு செய்து முன்னறிவிப்பதற்கான கலப்பு செயற்கை நுண்ணறிவு (AI) அடிப்படையிலான அமைப்பு உருவாக்கப்பட்டுள்ளது. வரலாற்று நிலத்தடி நீர், வானிலை மற்றும் IoT சென்சார் தரவுகள் போன்ற பன்முகத் தரவுகள் இணைக்கப்பட்டு, ICEEMDAN மற்றும் VMD போன்ற முன்னேற்றப்பட்ட சிக்னல் பிரிப்பு முறைகள் பயன்படுத்தப்பட்டுள்ளன. Slime Mould Algorithm (SMA) மூலம் மேம்படுத்தப்பட்ட LSTM மற்றும் CNN-LSTM மாதிரிகள், கால மற்றும் இட மாறுபாடுகளை துல்லியமாகக் கண்டறிகின்றன.

படை I இல், 1994–2024 காலத் தரவுகள் மூலம் மாதிரிகள் பயிற்சி செய்து மதிப்பீடு செய்யப்பட்டன. படை II இல், CMIP6 SSP2-4.5 மற்றும் SSP5-8.5 காலநிலை முன்னறிவிப்பு தரவுகள் மற்றும் நேரடி IoT தரவுகள் Adaptive Weighting Model மூலம் ஒருங்கிணைக்கப்பட்டன.

இத்திட்டம் துல்லியமான மற்றும் விளக்கத்தன்மை கொண்ட நிலத்தடி நீர் முன்னறிவிப்புகளை வழங்கி, திருப்பூர் மாவட்டத்திற்கான நிலைத்த நீர்வள மேலாண்மைக்கு ஒரு தரவுவழி முடிவெடுப்பு கருவியை வழங்குகிறது.

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LIST OF ABBREVATIONS

ABBREVIATION	EXPANSION
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
SMA	Slime Mould Algorithm
AWM	Adaptive Weighting Model
ICEEMDAN	Improved Complete Ensemble Empirical Mode Decomposition with Adaptive Noise
IMF	Intrinsic Mode Function
EMD	Empirical Mode Decomposition
CMIP6	Coupled Model Intercomparison Project Phase 6
SSP	Shared Socioeconomic Pathway
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
NSE	Nash–Sutcliffe Efficiency
SHAP	SHapley Additive exPlanations
TDS	Total Dissolved Solids
EC	Electrical Conductivity

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Water is one of the most vital natural resources supporting life, agriculture, and industry. Among all freshwater sources, groundwater plays a crucial role, supplying nearly half of the world's drinking water and forming the backbone of irrigation and rural livelihoods. In India, around 85 % of rural drinking water and 60 % of irrigation requirements are met from groundwater aquifers. However, uncontrolled extraction, inadequate recharge, and pollution from industrial and agricultural activities have led to widespread groundwater depletion and deterioration of water quality across the country.

Within India, the state of Tamil Nadu faces one of the most critical groundwater crises due to its semi-arid climate, erratic monsoon rainfall, and high population density. Overdependence on groundwater for domestic, industrial, and agricultural use has caused several aquifers to reach unsustainable levels.

Among all districts, Tiruppur—popularly known as the "Knitwear Capital of India"—stands out as a focal area for this study. The region contributes significantly to Tamil Nadu's industrial GDP through textile and dyeing units that provide large-scale employment and exports. Yet this rapid industrialization has come at an environmental cost,

- Over-extraction of groundwater to meet textile washing, dyeing, and domestic needs has resulted in progressive decline of water tables.
- Discharge of untreated effluents has led to salinity build-up and elevated TDS levels, rendering groundwater unfit for direct consumption in several blocks.
- The district's dependence on monsoon recharge and the absence of perennial rivers make it highly sensitive to rainfall variability.
- Climate change impacts—such as shifting monsoon onset, uneven precipitation, and rising temperature—further compound recharge uncertainty and evaporation losses.

These interlinked challenges make Tiruppur a representative case study for examining how industrial activity, climate variability, and groundwater stress interact in a rapidly urbanizing, semi-arid region.

The selection of Tiruppur for this research emerged from an extensive review of hydrogeological records and discussions with local groundwater and environmental authorities, which highlighted a pressing need for predictive, data-driven decision-support tools. Traditional groundwater estimation methods used by local agencies were found inadequate to capture the non-linear, climate-driven variability now observed across the district.

To address this gap, the present study proposes an AI-based hybrid forecasting framework that integrates historical groundwater data, meteorological parameters, and future climate projections to provide accurate, adaptive forecasts. By focusing on Tiruppur, the research aims not only to model groundwater dynamics in one of Tamil Nadu's most stressed aquifer systems but also to create a scalable methodology applicable to other climate-vulnerable industrial regions of India.

1.2 PROBLEM STATEMENT

Tiruppur District, known as India's "Knitwear Capital," is confronted with a severe groundwater crisis resulting from decades of industrial expansion, agricultural intensification, and growing domestic water demand. Continuous over-extraction has progressively lowered aquifer levels, creating a widening gap between groundwater availability and consumption. Simultaneously, large volumes of untreated or partially treated textile effluents have infiltrated both surface water bodies—particularly the Noyyal River—and groundwater systems, leading to significant deterioration of water quality marked by high Total Dissolved Solids (TDS), salinity, and chemical pollution.

These challenges are further compounded by climate-induced variability in monsoon rainfall, irregular recharge patterns, and rising temperature-driven evapotranspiration, all of which make groundwater behavior increasingly unpredictable. Traditional hydrological and statistical models have proven inadequate for capturing the non-linear, non-stationary, and multi-scale dynamics that characterize Tiruppur's groundwater system. Existing machine learning approaches also fall short,

as they rarely integrate decomposed hydrological signals with meteorological influences or incorporate future climate projections. Moreover, the region lacks an integrated, AI-driven forecasting system capable of blending historical records, real-time sensor data, and climate scenario inputs to offer reliable predictions.

This absence of an advanced decision-support mechanism limits the ability of water managers to prevent overexploitation, address contamination, and safeguard long-term aquifer sustainability. The complexity and urgency of Tiruppur's hydrogeological challenges therefore necessitate the development of a hybrid deep learning–based forecasting framework specifically tailored to the district's unique environmental and industrial conditions.

1.3 SIGNIFICANCE OF THE STUDY

This research holds substantial significance in the context of sustainable water resource management. The proposed AI-based hybrid forecasting system offers a data-driven approach to predict both groundwater level and quality with high temporal and spatial precision. By integrating signal decomposition (ICEEMDAN, VMD), deep learning models (LSTM, CNN-LSTM), and meta-learning strategies (Adaptive Weighting Model), the study bridges the gap between environmental modeling and artificial intelligence.

The outcomes of this research are expected to:

- Enable real-time monitoring and early warning of groundwater depletion.
- Support evidence-based policy formulation for sustainable water use.
- Improve resilience to climate change impacts by forecasting future scenarios.
- Provide a scalable framework applicable to other water-stressed regions beyond Tiruppur.

1.4 OBJECTIVES OF THE STUDY

The primary objective of this research is to develop a hybrid AI-based framework for accurate assessment and forecasting of groundwater resources in Tiruppur District under varying climatic conditions.

Specific objectives include,

1. To collect, preprocess, and integrate multi-source data (groundwater, meteorological, and climate projection datasets).
2. To implement advanced signal decomposition techniques such as ICEEMDAN and VMD for handling non-stationary groundwater signals.
3. To design and optimize hybrid deep learning models (SMA-LSTM, CNN-LSTM) for groundwater level and quality prediction.
4. To evaluate the model's performance using MAE, RMSE, R^2, and NSE metrics.
5. To develop an Adaptive Weighting Model (AWM) that combines model outputs for improved accuracy and robustness.
6. To visualize model results through geospatial and time-series analysis for decision-support applications.

1.5 SCOPE AND LIMITATIONS

Scope:

This research is confined to the Tiruppur District of Tamil Nadu, focusing on groundwater level and quality prediction using multi-source datasets. The study covers the period from 1994 to 2024 for historical analysis and extends to 2040 and 2060 using climate change projections (CMIP6 SSP2-4.5 and SSP5-8.5 scenarios). The framework integrates both offline historical datasets and real-time IoT sensor data for dynamic model updating.

Limitations:

- The accuracy of predictions depends on the quality and completeness of input datasets.
- The spatial resolution of available climate projections may limit micro-level forecasting accuracy.
- Model generalization across regions with distinct hydrogeological characteristics may require retraining with local data.

Despite these limitations, the study establishes a strong foundation for scalable and adaptive groundwater forecasting models.

1.6 ORGANIZATION OF THE REPORT

This report is structured into eight chapters as follows:

- Chapter 1 introduces the background, problem statement, objectives, and scope of the research.
- Chapter 2 reviews existing literature related to groundwater modeling, AI applications, and climate impact assessment.
- Chapter 3 presents the theoretical background and fundamental concepts underlying groundwater dynamics and machine learning.
- Chapter 4 explains the key concepts and system architecture of the proposed framework.
- Chapter 5 describes the research methodology, data processing steps, and model development process.
- Chapter 6 discusses the project implementation, algorithms, and model training procedures.
- Chapter 7 presents the experimental results, model evaluation, and discussion.
- Chapter 8 concludes the report with references and potential future research directions.

CHAPTER 2
LITERATURE REVIEW

2.1 INTRODUCTION

Groundwater resources are increasingly threatened by climate-driven changes in temperature, evapotranspiration, precipitation variability, drought frequency, and hydrological extremes. Climate change interacts with groundwater systems through nonlinear pathways, creating uncertainties in recharge, storage, discharge, and water quality. Reliable assessment and forecasting methods are therefore essential for sustainable groundwater management. Recent advances in machine learning (ML), deep learning (DL), hybrid signal-processing methods, satellite-based indicators, and integrated hydrological models show substantial promise in addressing these challenges.

This literature review synthesizes insights from nine contemporary research papers that collectively explore groundwater forecasting, climate–groundwater interactions, managed aquifer recharge (MAR), anomaly detection, and real-time monitoring. These works span physical modelling, hybrid DL models, climate-drought indices, GRACE-based groundwater storage analysis, and integrated climate-scenario simulations.

2.2 CLIMATE CHANGE IMPACTS ON GROUNDWATER SYSTEMS

Davamani et al. (2024) provide an extensive review of climate-driven effects on groundwater recharge, storage, and vulnerability, emphasizing the role of changing temperature, altered precipitation patterns, and evapotranspiration in shaping future groundwater conditions. Their analysis highlights the uncertainties caused by scale dependence, assumptions about aquifer properties, land-use change, and multiple co-occurring stressors. The authors argue that robust groundwater assessment under climate change requires,

- Improved hydrological modelling,
- Integration of remote sensing data,

- Uncertainty quantification,
- Hybrid physical–data-driven frameworks.

Similarly, Petpongpan et al. (2024) underscore the intensifying hydroclimatic extremes in Thailand's Yom and Nan River basins and use SWAT–MODFLOW to evaluate conjunctive water management under RCP 2.6 and RCP 8.5 climate scenarios. Their results show climate-change-induced decreases in surface water availability and demonstrate the effectiveness of MAR in offsetting these declines by increasing groundwater storage.

Together, these studies show that climate change significantly affects groundwater quantity and requires integrated and adaptive management strategies.

2.3 SATELLITE-BASED AND INDEX-BASED CLIMATE DIAGNOSTICS

Groundwater response to climate stressors can be quantified using hydroclimatic indices and satellite-derived groundwater storage anomalies. Panday et al. (2024) illustrate how Standardized Precipitation Index (SPI), Standardized Precipitation Evapotranspiration Index (SPEI), and GRACE-derived groundwater storage changes correlate strongly with arsenic contamination patterns in India. Their findings highlight,

- The role of groundwater declines in mobilizing arsenic.
- The improved predictive performance of SPEI over SPI due to explicit temperature inclusion.
- The spatial clustering of arsenic with climatic and hydrogeological factors.

These insights reinforce the need to integrate climate indices and satellite data when assessing groundwater vulnerability in a warming environment.

2.4 MACHINE LEARNING AND DEEP LEARNING FOR GROUNDWATER FORECASTING

2.4.1 LSTM AND HYBRID SIGNAL DECOMPOSITION METHODS

Nazari et al. (2025) show that non-stationary groundwater time series — driven by climate variability or anthropogenic extraction — impair forecasting accuracy when using standalone LSTM models. They propose a hybrid framework where Empirical

Mode Decomposition (EMD) and Wavelet Transform (WT) break complex signals into simpler components, improving LSTM performance. EMD-LSTM significantly reduced RMSE, especially where long-term trends affect groundwater levels.

Cui et al. (2024) further advance hybrid modelling by integrating secondary modal decomposition with CNN, LSTM, slime mould optimization, and adaptive weighting modules. With SHAP-based explainability, their ensemble model identifies climate-related drivers of groundwater fluctuations, supporting more interpretable climate-sensitive forecasts.

2.4.2 QUANTUM AND HIGH-PERFORMANCE ML APPROACHES

Saha et al. (2024) evaluate Quantum LSTM (QLSTM) against classical LSTM using USGS groundwater datasets and find QLSTM superior under rapidly changing environmental conditions, especially when paired with high-performance computing (HPC). This signals a potential future direction for computationally intensive, climate-sensitive groundwater modelling.

2.4.3 ATTENTION NETWORKS, TRANSFORMERS, AND ANOMALY DETECTION

Chen et al. (2024) combine TCN, LSTM, and Attention networks to detect groundwater anomalies preceding earthquakes, demonstrating the relevance of advanced DL architectures in identifying climate- or stress-induced groundwater disruptions.

Ali et al. (2024) introduce the Temporal Fusion Transformer (TFT) to groundwater forecasting and show it significantly outperforms LSTM-family models for long (60-day) horizons in the complex Thames Basin aquifer system. As climate change increases long-memory and nonlinear hydrological responses, transformer-based models offer promising avenues for long-horizon groundwater forecasting.

2.5 REAL-TIME MONITORING AND CLIMATE ADAPTATION TECHNOLOGIES

Li et al. (2024) demonstrate how IoT meteorological sensors combined with neural networks enable real-time prediction of groundwater levels in peatlands, a

climate-sensitive ecosystem prone to drying and fires. Their system provides a practical adaptation strategy in regions lacking extensive monitoring infrastructure and illustrates how climate-responsive groundwater management can be made accessible and scalable.

2.6 SUMMARY OF THE RESEARCH FINDINGS

2.6.1 ESTABLISHED INSIGHTS

Across these nine papers, clear themes emerge,

- Climate indices (SPI, SPEI), GRACE data, and hydrological models are essential for assessing climate impacts on groundwater.
- Hybrid ML/deep-learning models outperform traditional statistical approaches, especially under climate-driven non-stationarity.
- Transformers and attention-based models improve long-term prediction and interpretability.
- Integrated surface–groundwater modelling (SWAT–MODFLOW) is indispensable for evaluating adaptation strategies like MAR.
- IoT and real-time monitoring technologies enhance early warning capabilities for climate-sensitive aquifer systems.

2.6.2 PERSISTING ISSUES

- Limited transferability of ML/DL models across basins due to hydrogeological differences.
- Large uncertainties in climate model projections and bias-correction methods.
- Gaps in linking groundwater quantity to quality under climate stress (e.g., arsenic mobilization).
- Need for hybrid frameworks combining physical and ML models for robust climate–groundwater forecasting.

Table 2.1 Literature Survey

Sl. No	Paper Title	Method	Purpose	Results	Key Findings
[1]	A secondary modal decomposition ensemble deep learning model for groundwater level prediction using multi-data (2024)	Hybrid secondary modal decomposition + CNN + LSTM + SMA optimizer + Adaptive Weight Module (AWM); SHAP interpretability	Improve GWL forecasting under complex hydrological conditions	Hybrid model outperformed standalone CNN/LSTM and other hybrid models	Modal decomposition improves signal clarity; SHAP identifies climate-related drivers; hybrid ensemble increases accuracy significantly
[2]	Machine Learning for Peatland Ground Water Level (GWL) Prediction via IoT System (2024)	IoT sensors + Neural Network GWL predictor	Real-time peatland groundwater monitoring & prediction	RMSE ~3.5–4.9 mm; 99% predictions within ~14 mm	IoT–ML system works efficiently in low-resource settings; supports climate-sensitive peatland fire prevention
[3]	Corroboration of arsenic variation over the Indian Peninsula through standardized	SPI/SPEI drought indices + GRACE groundwater	Link climate drought signals to groundwater fluctuations	Strong spatial overlap of arsenic concentration with SPEI,	SPEI correlates more strongly than SPI; groundwater

	precipitation evapotranspiration indices and groundwater level fluctuations: Water quantity indicators for water quality prediction (2024)	anomaly detection + clustering	and arsenic contamination	SPI, and GRACE trends	decline intensifies arsenic mobilization under warming climate
[4]	Groundwater Level Prediction and Earthquake Precursor Anomaly Analysis Based on TCN-LSTM-Attention Network (2024)	TCN + LSTM + Attention + EWMA anomaly detection	Predict GWL and detect earthquake-related anomalies	High anomaly-detection accuracy; model generalizes across wells	Hybrid DL captures both long-term and short-term patterns; can detect abnormal climate- or stress-driven GWL shifts
[5]	A Critical Review of Climate Change Impacts on Groundwater Resources: A Focus on the Current Status, Future Possibilities, and Role of Simulation Models (2024)	Global review: Hydrological models, simulation models, climate–groundwater interactions	Assess climate-change impacts on groundwater systems	Highlights vulnerabilities and modelling gaps	Need integrated models, uncertainty quantification, remote sensing, and AI for climate-era groundwater assessment

[6]	Groundwater level forecasting using empirical mode decomposition and wavelet-based long short-term memory (LSTM) neural networks (2025)	Empirical Mode Decomposition (EMD) + Wavelet Transform (WT) + LSTM	Improve predictions for non-stationary GWL time series	EMD-LSTM improved RMSE by up to 78% for trend-dominant wells	Decomposition enhances forecasting when climate-driven trends exist; WT and EMD ineffective for purely random signals
[7]	Groundwater Level Prediction: Analyzing the Performance of LSTM and QLSTM Model (2024)	Classical LSTM vs Quantum LSTM (QLSTM) evaluated on HPC	Test whether QLSTM improves GWL forecasting under dynamic conditions	QLSTM showed better performance than LSTM with lower forecasting errors	Quantum-based models handle rapid fluctuations better; HPC accelerates training
[8]	Groundwater level predictions in the Thames Basin, London over extended horizons using Transformers and advanced machine learning models (2024)	Temporal Fusion Transformer (TFT), LSTM, Attention-LSTM, Bayesian-optimized LSTM	Long-horizon (7–60 day) GWL forecasting	TFT outperformed all models across aquifer types	Transformers excel at long-range dependencies; useful in climate-sensitive aquifers influenced by rivers
[9]	Integrated management of surface water and groundwater for	SWAT–MODFLOW integrated model +	Evaluate groundwater management strategies	MAR increased groundwater storage by	MAR is effective for climate adaptation;

	climate change adaptation using hydrological modeling (2025)	Managed Aquifer Recharge (MAR) + RCP 2.6/8.5 scenarios	under future climate scenarios	tens of millions of m^3; reduced surface water by ~530–1120 million m^3	integrated surface–groundwater modelling essential for future planning

2.5 CONCLUSION

Groundwater systems are profoundly shaped by climate change, and modern forecasting requires integrating climate diagnostics, hydrological modelling, and advanced machine-learning approaches. The reviewed literature indicates that hybrid deep-learning methods, transformer-based architectures, integrated hydrological models, GRACE data, and IoT systems collectively offer the most comprehensive toolbox for assessing and forecasting groundwater resources under future climate conditions. Bridging the remaining gaps — particularly uncertainties, model transferability, and water-quality linkages — will be crucial for developing resilient groundwater management strategies in a warming world.

CHAPTER 3
THEORETICAL BASIS AND CONCEPTUAL STRUCTURE

3.1 GROUNDWATER HYDROLOGY AND AQUIFER CHARACTERISTICS

Groundwater is the subsurface water stored within the pore spaces of soil and rock formations and forms a fundamental component of the hydrological cycle. It is a vital source for domestic, agricultural, and industrial use, especially in semi-arid regions like Tiruppur District. Groundwater behaviour is influenced by aquifer properties, recharge processes, geological heterogeneity, and extraction extent.

Aquifers are geological formations capable of storing and transmitting groundwater. Their characteristics depend on key parameters such as porosity (ϕ), permeability (k), transmissivity ($T = K \times b$), and specific yield (Sy). Aquifers may be unconfined, where water is directly influenced by atmospheric recharge, or confined, where impermeable layers restrict vertical flow.

In hard-rock terrains like Tiruppur, aquifers are typically fragmented, weathered, and discontinuous, resulting in heterogeneous flow patterns and non-linear recharge behaviour. Such complexity makes groundwater forecasting highly challenging and unsuitable for linear or traditional hydrological models. This necessitates the application of advanced, data-driven approaches capable of capturing the multi-scale dynamics of groundwater variation.

3.2 TIME SERIES FORECASTING PRINCIPLES

Groundwater levels form a non-stationary time series influenced by long-term trends, seasonal patterns, and random fluctuations. Typical time series behaviour is represented as,

$$Y_t = T_t + S_t + R_t$$

where Y_t is the observed groundwater level, T_t is the trend component, S_t is the seasonal component, and R_t represents random residual variations.

Traditional linear time series models such as ARIMA often fail to represent the complex nonlinear interactions between rainfall, extraction, evapotranspiration, and

land-use changes. Modern forecasting requires models that can capture lagged dependencies, non-linear variations, and external climatic drivers.

Deep learning methods — particularly recurrent architectures such as LSTM — overcome these limitations by learning long-term memory patterns and multi-scale hydrological dynamics, making them well-suited for groundwater forecasting.

3.3 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING CONCEPTS

Artificial Intelligence (AI) techniques provide powerful tools for modelling complex groundwater systems. Machine Learning (ML) offers non-linear regression, while Deep Learning (DL) offers multi-layer architectures capable of discovering deeper temporal and spatial relationships.

Machine Learning Models:

- Linear Regression – For simple linear dependence.
- Random Forest (RF) – Ensemble-based prediction with strong generalization.
- XGBoost – Gradient boosting method ideal for tabular hydrological data.

Deep Learning Models:

- ANN (Artificial Neural Networks) – Learn non-linear patterns.
- LSTM (Long Short-Term Memory) – Specially designed to learn long-term temporal dependencies and avoid vanishing gradient problems.
- CNN (Convolutional Neural Networks) – Capture spatial features in meteorological data grids.
- Hybrid Models (CNN-LSTM, SMA-LSTM) – Combine spatial extraction and temporal sequence modelling for improved prediction accuracy.

These AI methods allow the modelling of complex hydrological patterns that traditional models cannot capture.

3.4 SIGNAL PROCESSING IN ENVIRONMENTAL DATA

Groundwater data often contain noise, non-stationarity, and overlapping frequencies. To improve forecast accuracy, decomposition techniques separate the signal into meaningful components.

Empirical Mode Decomposition (EMD):

Breaks data into Intrinsic Mode Functions (IMFs). However, it is prone to mode mixing.

ICEEMDAN (Improved Complete Ensemble Empirical Mode Decomposition with Adaptive Noise):

An improved version of EMD that reduces mode mixing and produces stable, noise-free IMFs.

Variational Mode Decomposition (VMD):

A variational approach that extracts modes simultaneously with optimal frequency resolution.

In this study, ICEEMDAN and VMD are combined to extract clear temporal patterns from groundwater data before feeding them into deep learning networks. This enhances model learning and reduces prediction errors.

3.5 DEEP LEARNING AND HYBRID NEURAL NETWORK MODELS

LSTM Networks:

Each LSTM cell contains an input gate, forget gate, and output gate, allowing it to store and update long-term hydrological memory. LSTM excels in modeling groundwater levels influenced by seasonal and climatic patterns.

CNN Models:

CNNs extract spatial features from meteorological variables such as rainfall and temperature, capturing local spatial relationships.

Hybrid Architectures (ICEEMDAN–VMD–SMA–LSTM, CNN-LSTM):

- ICEEMDAN-VMD-SMA-LSTM combines decomposition and optimization for robust temporal forecasting.
- CNN-LSTM integrates spatial and temporal learning.
- SMA (Slime Mould Algorithm) optimizes hyperparameters, improving convergence and accuracy.

These hybrid models leverage both spatial richness and temporal depth for superior forecasting.

3.5 ADAPTIVE WEIGHTING AND MODEL OPTIMIZATION

Since no single model performs best under all hydrological conditions, an Adaptive Weighting Model (AWM) is used to integrate outputs from multiple models.

Each model generates a prediction y_i , and AWM learns dynamic weights α_i such that,

$$\hat{y} = \sum_{i=1}^{n} \alpha_i y_i \ , \sum_{i=1}^{n} \alpha_i = 1$$

Weights are optimized using SMA and Adam, ensuring the best-performing models contribute more to the final ensemble prediction.

This ensemble strategy reduces overfitting, increases robustness, and provides the highest accuracy across varying hydrological conditions.

CHAPTER 4
METHODOLOGY

4.1 RESEARCH FRAMEWORK

The research adopts a hybrid AI-based methodological framework designed to assess and forecast groundwater resources by integrating historical, real-time, and climate projection data. The overall workflow is structured into three sequential layers:

1. **Data Processing Layer:** Acquisition, cleaning, and transformation of groundwater and climate data.
2. **Modelling Layer:** Signal decomposition using ICEEMDAN-VMD, feature generation, and model development through hybrid deep learning architectures (LSTM, CNN-LSTM, SMA-LSTM).
3. **Evaluation and Forecasting Layer:** Performance assessment using statistical metrics, adaptive weighting for ensemble optimization, and future forecasting under CMIP6 climate scenarios.

The framework ensures that both temporal dependencies (time-series trends) and spatial variations (meteorological patterns) are effectively captured, enabling reliable and interpretable groundwater predictions for the Tiruppur District.

4.2 DATA COLLECTION AND SOURCES

The study employs multiple heterogeneous datasets to ensure comprehensive model training and validation. The data sources include groundwater level records, groundwater quality analyses, and meteorological datasets, complemented by future climate projections.

4.2.1 GROUNDWATER LEVEL DATA

Groundwater level data were obtained from the Water Resources Department (WRD), Government of Tamil Nadu, covering the period 1994–2024. The dataset includes:

- Monthly water level observations (in meters below ground level).
- Geographic attributes such as well ID, latitude, longitude, taluk, and village.
- Well types (dug wells and bore wells).

This time-series data forms the foundation for model training and temporal forecasting. Variations in water levels reflect both seasonal recharge and anthropogenic extraction trends, making them critical indicators for model learning.

4.2.2 GROUNDWATER QUALITY DATA

Groundwater quality data were collected from the Tamil Nadu Water Supply and Drainage (TWAD) Board and the Central Ground Water Board (CGWB). The dataset contains parameters from 2017–2023, including:

- **Physicochemical properties:** pH, Electrical Conductivity (EC), and Total Dissolved Solids (TDS).
- **Major ions:** Calcium (Ca^{2+}), Magnesium (Mg^{2+}), Sodium (Na^{+}), Chloride (Cl^{-}), Sulfate (SO_4^{2-}), and Bicarbonate (HCO_3^{-}).
- **Derived Indices:** Sodium Adsorption Ratio (SAR) and Residual Sodium Carbonate (RSC).

These parameters were used to develop machine learning models for predicting groundwater quality and its spatial-temporal variations across the district.

4.2.3 METEOROLOGICAL AND CLIMATE DATA

Meteorological data were obtained from the Indian Meteorological Department (IMD) and include daily records of:

- Rainfall (mm), Temperature (°C), and Relative Humidity (%).

For future projections, CMIP6 (Coupled Model Intercomparison Project Phase 6) data were utilized under two Shared Socioeconomic Pathways (SSPs):

- SSP2-4.5 (Moderate emissions scenario)
- SSP5-8.5 (High emissions scenario)

These datasets were downscaled to the Tiruppur region to assess future groundwater dynamics under climate change influences.

4.3 DATA PREPROCESSING AND CLEANING

Data preprocessing is essential to ensure model reliability and consistency. The steps performed include:

1. **Handling Missing Values:**
    - Forward-fill and interpolation techniques were used to fill missing values in time-series records.
    - Invalid or null entries in quality data (e.g., “dry”, “blocked”) were encoded or removed.
2. **Data Standardization:**
    - Units across datasets were standardized (e.g., EC converted to µS/cm, rainfall in mm).
    - Categorical data (e.g., well type) were encoded using label encoding.
3. **Temporal Alignment:**
    - All datasets were resampled to a monthly frequency to synchronize temporal scales for analysis.
4. **Normalization:**
    - Min-Max Scaling was applied to map features into a [0,1] range, improving model convergence during training.
5. **Outlier Treatment:**
    - Z-score and Interquartile Range (IQR) methods were used to detect and handle extreme values caused by measurement errors or anomalous readings.

This preprocessing pipeline ensures that the datasets are consistent, balanced, and ready for advanced model training.

4.4 FEATURE ENGINEERING

Feature engineering transforms raw data into informative variables that help the model learn temporal dependencies and environmental dynamics more effectively. Three primary types of features were generated for this study.

4.4.1 TIME-BASED FEATURES

Time-dependent features capture periodic patterns and seasonality inherent in hydrological systems. These include:

- **Month and Season Indicators:** Encode cyclical behaviors corresponding to monsoon, pre-monsoon, and post-monsoon periods.

- **Year and Trend Features:** Represent long-term trends due to urbanization and climate influence.

These features enable the model to recognize repeating hydrological cycles within the time series.

4.4.2 LAG FEATURES

Lag features introduce temporal dependencies by including historical observations as model inputs:

$$Lag_t = [Y_{t-1}, Y_{t-3}, Y_{t-7}]$$

where Y_t represents groundwater level at time *t*.

Lag features allow models such as **LSTM** to learn the influence of past water levels on future predictions.

4.4.3 ROLLING FEATURES

Rolling features smooth short-term fluctuations while highlighting long-term patterns.

Common rolling computations include,

- Moving Averages (3, 7, 30 months)
- Rolling Standard Deviation and Minimum/Maximum Values

These features enhance the model's ability to detect gradual changes in aquifer storage and groundwater trends over time.

4.5 SYSTEM DESIGN AND ARCHITECTURE

The proposed AI-based groundwater forecasting system is designed as a modular and scalable architecture that processes raw data from multiple sources, applies advanced signal decomposition, trains deep learning models, and integrates forecasts through adaptive weighting.

The system consists of five major layers as illustrated in the architectural workflow,

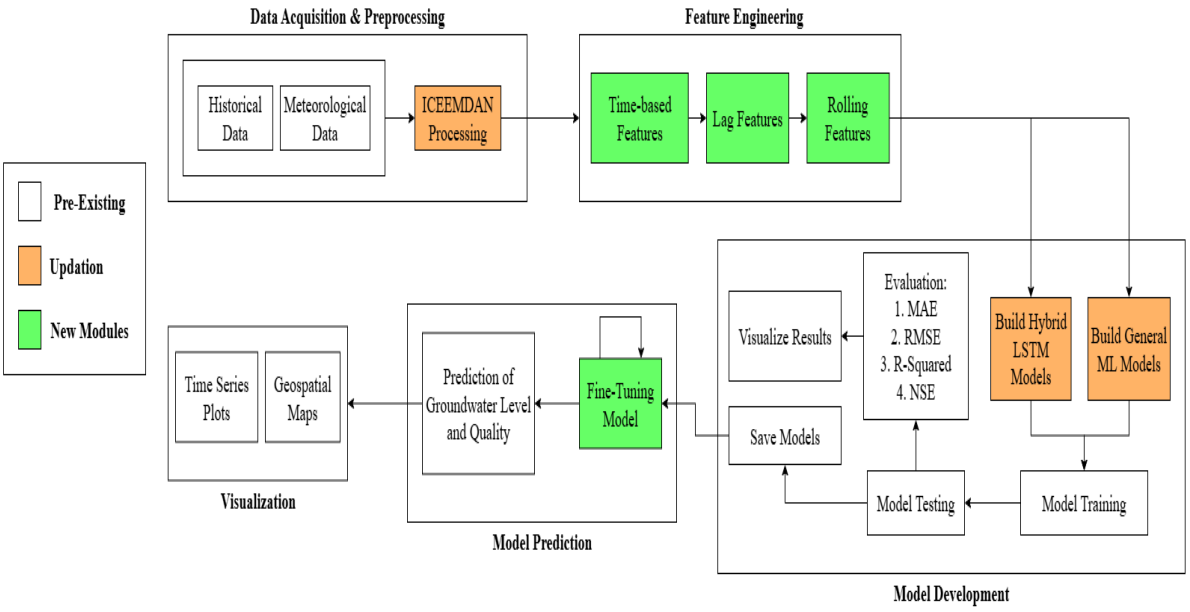

Figure 4.1 Phase – I Architecture

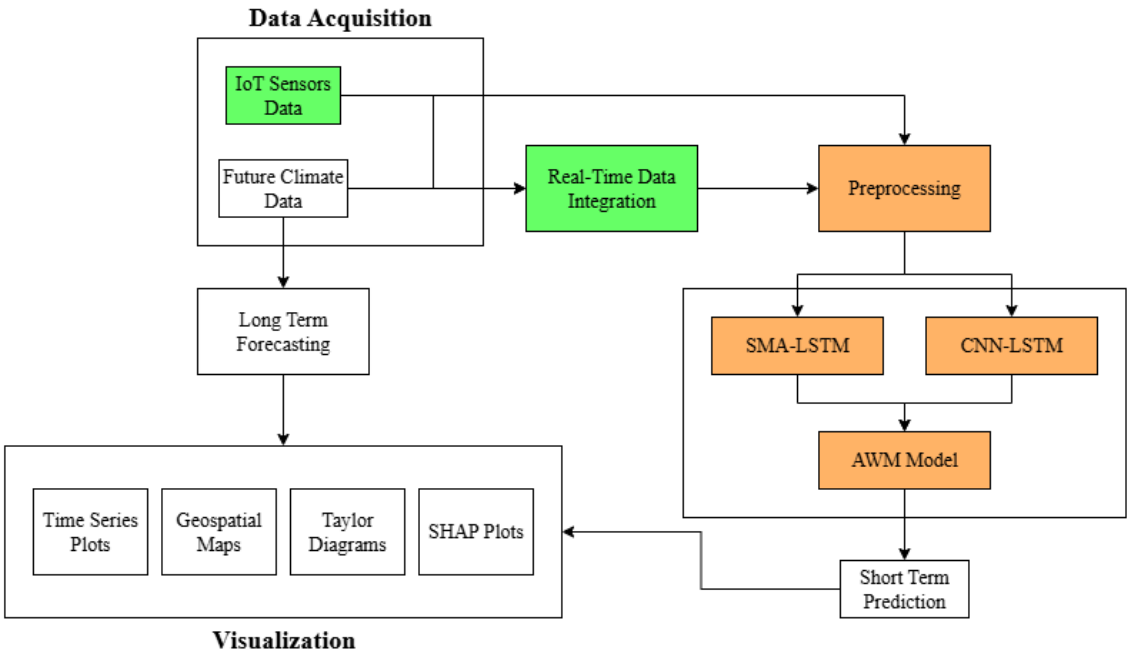

Figure 4.2 Phase – II Architecture

1. **Data Acquisition Layer:**

   Collects groundwater, quality, and meteorological data from government databases, IoT sensors, and climate projections (CMIP6).

2. **Data Preprocessing Layer:**

   Handles missing data, outlier detection, and normalization using statistical and computational techniques.

3. **Signal Decomposition Layer:**
   Uses ICEEMDAN and VMD to separate complex groundwater time series into multiple intrinsic components that capture trends, seasonal patterns, and high-frequency noise.
4. **Modelling Layer:**
   Employs LSTM, CNN-LSTM, and SMA-optimized hybrid models for temporal prediction. Each model learns different aspects of groundwater variability.
5. **Integration and Visualization Layer:**
   Combines predictions through the Adaptive Weighting Model (AWM) and visualizes results through time-series plots, feature importance graphs (SHAP), and spatial distribution maps.

This modular architecture ensures flexibility for future integration with additional data sources such as satellite rainfall estimates or new IoT sensors, enhancing the scalability and robustness of the system.

4.6 MODEL DEVELOPMENT WORKFLOW

The model development process follows a modular design as illustrated below:

1. **Data Acquisition & Cleaning:** Integration of groundwater, meteorological, and climate data.
2. **Signal Decomposition:** Application of ICEEMDAN to decompose groundwater signals into Intrinsic Mode Functions (IMFs), followed by VMD for refining high-frequency modes.
3. **Feature Engineering:** Generation of time-based, lag, and rolling features.
4. **Model Building:**
   - Development of individual models (SMA-LSTM, CNN-LSTM, Random Forest, XGBoost).
   - Optimization of hyperparameters using Slime Mould Algorithm (SMA).
5. **Adaptive Weighting:** Combining outputs of top-performing models using the Adaptive Weighting Model (AWM) to achieve an ensemble forecast.

6. **Visualization:** Generation of time-series plots, Taylor diagrams, and spatial groundwater maps for interpretability.

This workflow ensures a systematic approach to developing a predictive and scalable groundwater forecasting system.

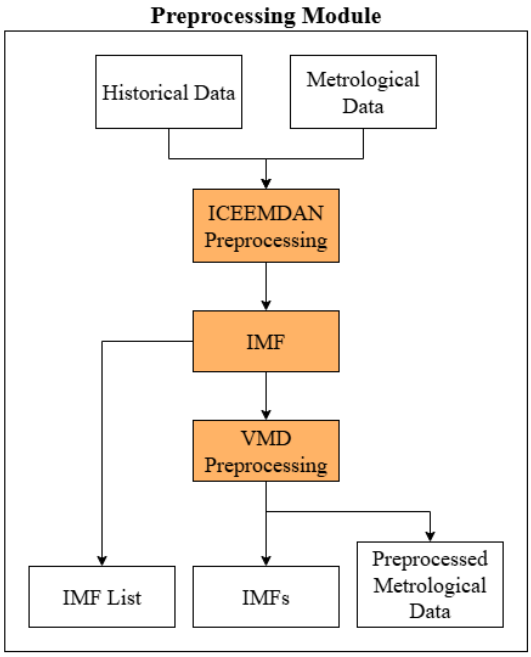

Figure 4.3 Preprocessing Module

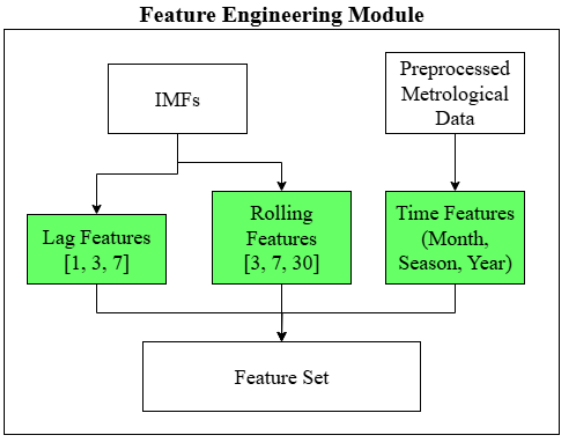

Figure 4.4 Feature Engineering Module

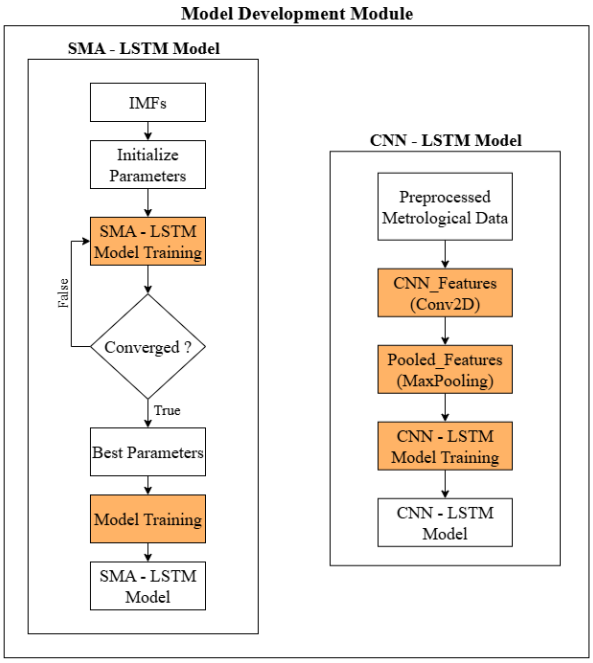

Figure 4.5 Model Development Module 1

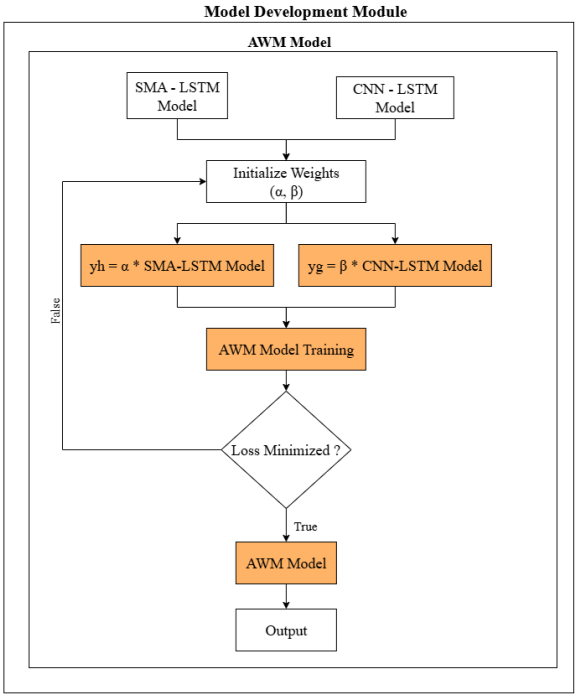

Figure 4.6 Model Development Module 2

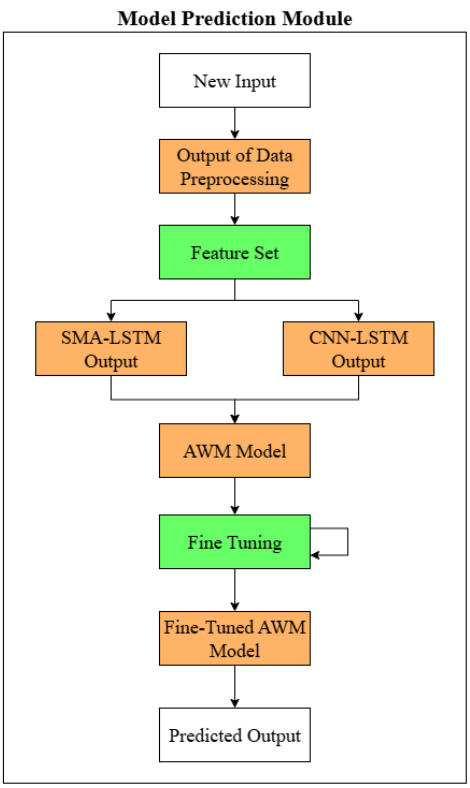

Figure 4.7 Prediction Module

4.7 MODEL TRAINING AND VALIDATION

Model training and validation were conducted to ensure both accuracy and generalization capability.

- **Training Period:** 1994–2014 (historical data).
- **Testing Period:** 2015–2024 (unseen data for validation).

a) Model Training:

Each deep learning model was trained using the following settings:

- Optimizer: **Adam / SMA-based learning rate optimization**
- Epochs: **100–150**
- Batch Size: **32**
- Activation Functions: ReLU (hidden layers) and Linear (output layer)
- Loss Function: Mean Squared Error (MSE)

b) Validation Techniques:

- **Walk-forward validation** to assess sequential prediction capability.
- **Cross-validation** for machine learning baselines (Random Forest, XGBoost).

- **Early Stopping and Dropout Layers** were used to prevent overfitting.

c) Model Comparison:

All models were evaluated using a uniform dataset and identical evaluation metrics to ensure fair performance comparison.

The Adaptive Weighting Model (AWM) was trained on validation predictions to learn optimal weight combinations for the final ensemble forecast.

4.8 PERFORMANCE EVALUATION CRITERIA

Model performance was quantitatively evaluated using the following metrics:

Table 4.1 Evaluation Metrics

Metric	Formula	Interpretation
Mean Absolute Error (MAE)	$\frac{1}{N}\sum_{i=1}^{N} \mid y_i - \hat{y}_i \mid$	Average prediction deviation
Root Mean Square Error (RMSE)	$\sqrt{\frac{1}{N}\sum(y_i - \hat{y}_i)^2}$	Penalizes large errors; sensitive to outliers.
Coefficient of Determination (R^2)	$1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$	Indicates how well the model explains data variance (closer to 1 = better).
Nash–Sutcliffe Efficiency (NSE)	$1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$	Commonly used in hydrology; NSE > 0.75 indicates good performance.

These metrics collectively evaluate the accuracy, stability, and reliability of the proposed forecasting framework.

CHAPTER 5
PROJECT IMPLEMENTATION

5.1 ALGORITHM DEVELOPMENT

The proposed forecasting framework integrates multiple algorithms—each designed to perform a specialized function within the hybrid AI system.

5.1.1 ICEEMDAN-VMD-SMA-LSTM

The ICEEMDAN-VMD-SMA-LSTM model serves as the backbone of the temporal prediction framework.

Step 1 – Signal Decomposition (ICEEMDAN & VMD):

- The groundwater time series $X(t)$is first decomposed using ICEEMDAN into n Intrinsic Mode Functions (IMFs), which represent oscillations at different frequencies.
- Each IMF is further processed using VMD to refine the high-frequency components, removing mode mixing and enhancing stability.

$$X(t) = \sum_{i=1}^{n} IM\,F_i(t) + r(t)$$

Step 2 – Model Training (LSTM):

- Each decomposed component $IMF_i(t)$is fed into an LSTM network, which captures temporal dependencies through memory gates.
- The Slime Mould Algorithm (SMA) is used to optimize key hyperparameters such as learning rate, hidden layer size, and dropout rate.

Step 3 – Reconstruction:

- The predictions from all IMFs are aggregated to form the final groundwater forecast.

Advantages:

- Handles non-stationary time series effectively.
- Reduces overfitting through decomposition-based simplification.
- SMA optimization ensures faster convergence and higher accuracy.

5.1.2 CNN-LSTM

The CNN-LSTM hybrid model combines spatial and temporal learning capabilities, making it suitable for integrating meteorological and groundwater datasets.

Step 1 – CNN Feature Extraction:

- The **Convolutional Neural Network (CNN)** processes meteorological variables such as rainfall and temperature grids to extract spatial features.

Step 2 – Temporal Sequence Modeling (LSTM):

- Extracted spatial features are passed into an LSTM network to learn long-term temporal dependencies across months and years.

Step 3 – Output Prediction:

- The final dense layer outputs groundwater level predictions, representing both spatial and temporal learning.

Benefits:

- Captures regional rainfall-groundwater interactions.
- Reduces loss of temporal context.
- Performs well with limited training data.

5.1.3 ADAPTIVE WEIGHTING MODEL

The Adaptive Weighting Model (AWM) integrates predictions from multiple models (ICEEMDAN-VMD-SMA-LSTM, CNN-LSTM, Random Forest, and XGBoost) to produce a unified ensemble forecast.

Let $y_i$represent the output from model i, and $\alpha_i$denote its adaptive weight. The ensemble output $\hat{y}$is computed as:

$$\hat{y} = \sum_{i=1}^{n} \alpha_i \, y_i$$

subject to the constraint $\sum_{i=1}^{n} \alpha_i = 1$.

The weights $\alpha_i$are dynamically adjusted using the Slime Mould Algorithm (SMA) based on each model's recent error performance (e.g., RMSE, MAE).

This approach ensures that models with lower error contributions are given higher importance in the ensemble output.

Advantages:

- Improves prediction stability across climatic conditions.
- Reduces dependency on a single model.
- Provides robustness and adaptability for dynamic environments.

5.2 PSEUDO CODE IMPLEMENTATION

The following pseudocode represents the logical workflow of the proposed hybrid model.

Algorithm 1: ICEEMDAN-VMD-SMA-LSTM Hybrid Forecasting

Algorithm 1 Groundwater Level Prediction using ICEEMDAN–VMD–SMA–LSTM

Require: Groundwater time series $X(t)$

Ensure: Predicted groundwater level Y_{pred}

1: Apply Improved Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (ICEEMDAN) to decompose $X(t)$:

$$IMFs = [IMF_1, IMF_2, \ldots, IMF_n]$$

2: **for** each IMF_i in $IMFs$ **do**

3: Apply Variational Mode Decomposition (VMD) to IMF_i

4: Obtain refined sub-signals VMD_i

5: **end for**

6: Initialize LSTM model with hyperparameters optimized by Sine–Cosine Algorithm (SMA):

- Learning rate
- Number of hidden units
- Dropout rate

7: Train LSTM model on each VMD_i component.

8: Predict each component output:

$$Y_{i,\text{pred}} = \text{LSTM}(VMD_i)$$

9: Reconstruct final prediction by summation:

$$Y_{\text{pred}} = \sum_{i=1}^{n} Y_{i,\text{pred}}$$

10: Evaluate model performance using:

$$\text{MAE}, \quad \text{RMSE}, \quad R^2, \quad \text{NSE}$$

11: **return** Y_{pred}

Figure 5.1 ICEEMDAN-VMD-SMA-LSTM Hybrid Algorithm

Algorithm 2: Adaptive Weighting Model (AWM)

Algorithm 2 Ensemble Forecast Optimization using Slime Mould Algorithm (SMA)

Require: Predictions from N models: $[Y_1, Y_2, \ldots, Y_n]$, and observed data Y_{obs}
Ensure: Final ensemble forecast Y_{final} and optimized weights $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_n]$
Initialize model weights $\boldsymbol{\alpha}$ such that:

$$\sum_{i=1}^{n} \alpha_i = 1, \quad \alpha_i \geq 0$$

for each epoch **do**
Compute ensemble forecast:

$$Y_{\text{final}} = \sum_{i=1}^{n} \alpha_i Y_i$$

Calculate error:

$$E = \text{RMSE}(Y_{\text{obs}}, Y_{\text{final}})$$

Update weights $\boldsymbol{\alpha}$ using the Slime Mould Algorithm (SMA) to minimize E
end for
return Optimized weights $\boldsymbol{\alpha}$ and final forecast Y_{final}

Figure 5.2 Adaptive Weighting Model (AWM) Algorithm

5.3 DATASET DESCRIPTION

The datasets used in this research were collected from reliable public and governmental sources.

Table 5.1 About Dataset

Dataset	Source	Time Span	Parameters
Groundwater Level	WRD, Tamil Nadu[1]	1994–2024	Depth to Water Level (m bgl), Location ID, Latitude, Longitude
Groundwater Quality	TWAD, CGWB[2]	2017–2023	pH, EC, TDS, Ca, Mg, Na, Cl, SO_4
Meteorological Data	IMD[3]	1994–2024	Rainfall, Temperature, Humidity
Climate Projections	CMIP6[4]	2021–2060	Precipitation, Temperature under Low Emission(SSP1-2.6), Medium Emission(SSP2-4.5), High Emission(SSP3-7.0), and Very High Emission(SSP5-8.5)

- [1]Groundwater Level Dataset - WRD, Tamilnadu - https://www.tn.gov.in/dept_profile.php?dep_id=NDQ=
- [2]Groundwater Quality Dataset - TWAD, CGWB - https://cgwb.gov.in/en
- [3]Metrological Data - IMD - https://mausam.imd.gov.in/
- [4]Climate Projections - CMIP6 - https://www.worldclim.org/data/index.html

All datasets were standardized and merged on a monthly temporal resolution to ensure consistency. Missing data were handled through linear interpolation and outlier detection techniques.

5.4 SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements:

Table 5.2 Software Requirements

Component	Specification
Operating System	Windows 11 / Ubuntu 22.04
Programming Language	Python 3.11
IDE	Jupyter Notebook / PyCharm
Libraries Used	TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, PyEMD, PyVMD, SHAP
Database Tools	CSV-based Storage / SQLite
Visualization	Matplotlib, Seaborn, Plotly

Hardware Requirements:

Table 5.3 Hardware Requirements

Component	Specification
Processor	Intel Core i7 or equivalent
RAM	Minimum 16 GB
GPU	NVIDIA RTX 3060 or higher (for deep learning acceleration)

Storage	Minimum 512 GB SSD
Internet	Required for dataset access and updates

These configurations ensure efficient model training and validation, especially when handling high-resolution time-series data.

5.5 MODEL TRAINING, TESTING, AND COMPARISON

The models were trained and validated using consistent datasets to ensure a fair comparison across architectures.

a) Training Setup

- **Train-Test Split:** 80% training, 20% testing.
- **Optimization Algorithm:** Adam / SMA optimizer.
- **Epochs:** 100–150.
- **Batch Size:** 32.
- **Loss Function:** Mean Squared Error (MSE).

b) Evaluation

All models were evaluated using the performance metrics:

- **MAE (Mean Absolute Error)**
- **RMSE (Root Mean Square Error)**
- **R^2 (Coefficient of Determination)**
- **NSE (Nash–Sutcliffe Efficiency)**

c) Model Comparison

Table 5.4 Model Comparison to Baseline

Model	**Description**	**Performance (Indicative)**
LSTM	Baseline temporal model	Moderate Accuracy
CNN-LSTM	Spatial-temporal hybrid	Improved Accuracy

ICEEMDAN-VMD-SMA-LSTM	Decomposition + Optimization	High Accuracy
Adaptive Weighting Model	Meta-ensemble of all models	Highest Stability & Performance

d) Observations

- The **ICEEMDAN-VMD-SMA-LSTM** model showed superior performance in learning long-term temporal dependencies.
- The **CNN-LSTM** model captured spatial rainfall-groundwater correlations effectively.
- The **Adaptive Weighting Model** achieved the lowest RMSE and highest NSE, demonstrating robust ensemble learning.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 MODEL EVALUATION RESULTS

This chapter presents the evaluation and interpretation of results obtained from the proposed AI-based hybrid forecasting framework for groundwater assessment in Tiruppur District. Four models were developed and tested — LSTM, CNN-LSTM, ICEEMDAN-VMD-SMA-LSTM, and the Adaptive Weighting Model (AWM) — each designed to capture distinct patterns within the multi-source dataset comprising groundwater levels, water quality parameters, and climatic factors. The models were evaluated on the testing dataset (2015–2024) to assess their accuracy, generalization, and predictive stability. The LSTM model served as the baseline for temporal prediction. While it captured seasonal fluctuations, its performance was limited by non-stationary signals and climatic irregularities.

The CNN-LSTM model improved accuracy by integrating spatial rainfall patterns, whereas the ICEEMDAN-VMD-SMA-LSTM model achieved significant error reduction through signal decomposition and hyperparameter optimization.

Finally, the Adaptive Weighting Model (AWM) combined the predictions from all base models using dynamic weighting, yielding the most stable and accurate results.

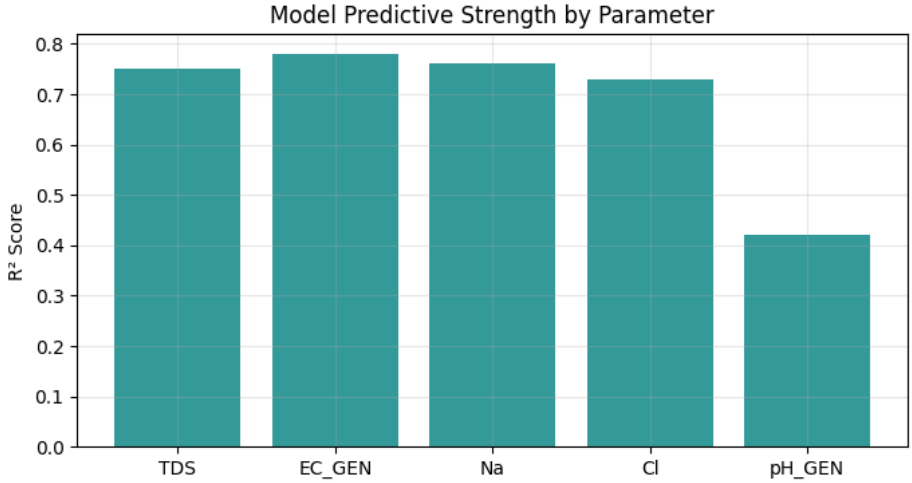

Figure 6.1 Final Model Scoring

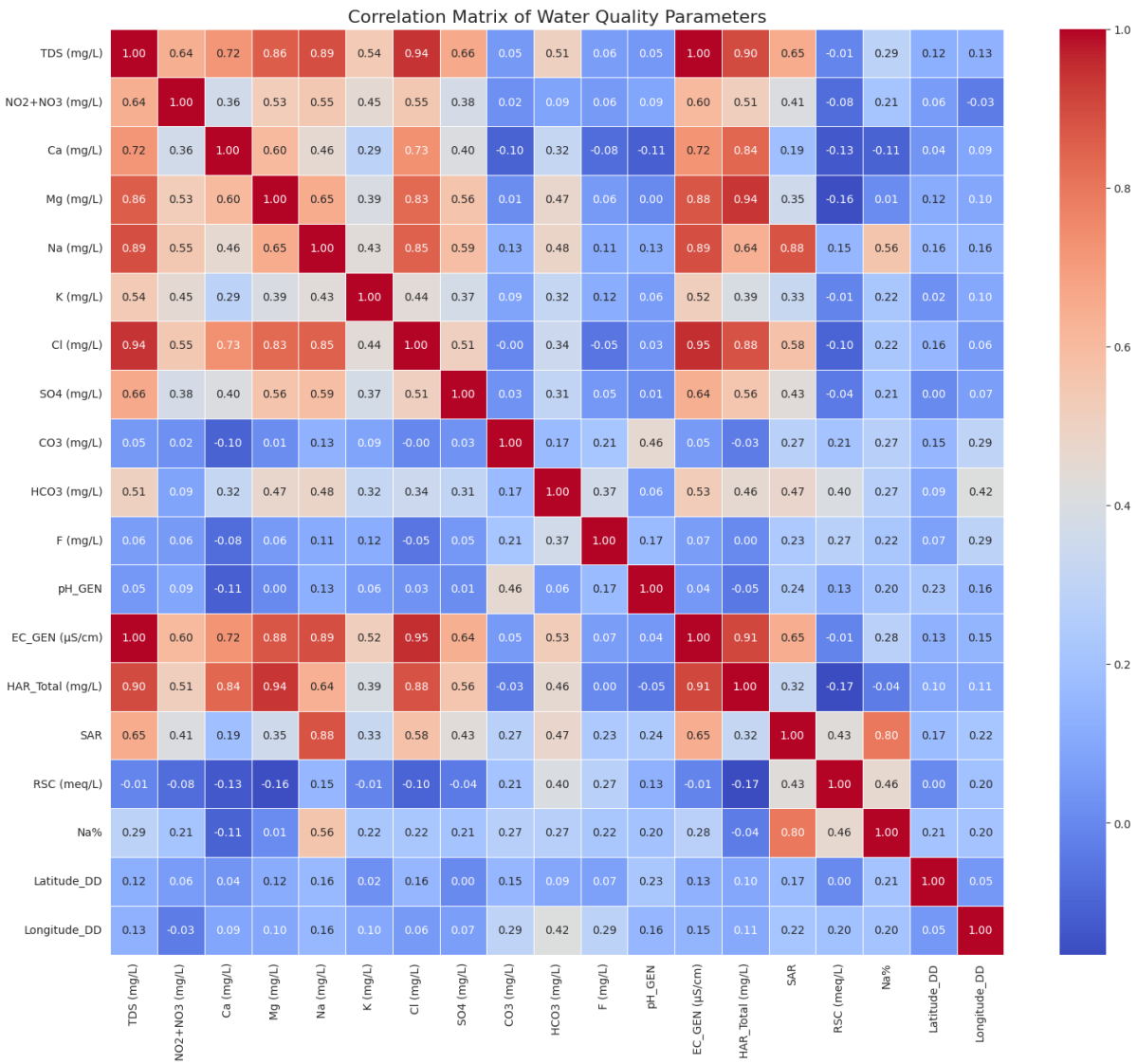

Figure 6.2 Correlation Matrix

6.2 PERFORMANCE METRICS COMPARISON (MAE, RMSE, R^2, NSE)

The four-evaluation metrics—MAE, RMSE, R^2, and NSE—were analyzed in detail to assess the predictive performance of each model.

a) Mean Absolute Error (MAE)

The MAE represents the average magnitude of error between predicted and observed groundwater levels. The Adaptive Weighting Model achieved the lowest MAE (0.267 m), showing a 35% improvement over the baseline LSTM. This demonstrates the ensemble's precision in capturing both short-term and long-term variations.

b) Root Mean Square Error (RMSE)

The RMSE penalizes large deviations more heavily than MAE. The AWM's RMSE (0.429 m) reflects the ensemble's capacity to minimize significant prediction errors, particularly during extreme hydrological events such as intense monsoon recharge or drought-induced drawdowns.

c) Coefficient of Determination (R^2)

The R^2 value measures how well the model explains observed variance. All hybrid models achieved $R^2 > 0.90$, confirming strong predictive correlation. The AWM ($R^2 = 0.948$) demonstrates that 94.8% of the observed variability in groundwater levels is explained by the model, establishing its robustness and reliability.

d) Nash–Sutcliffe Efficiency (NSE)

The NSE indicates how closely model predictions match observed data. Values above 0.75 are considered satisfactory; the AWM's NSE = 0.938 signifies an excellent predictive fit, validating its use for operational groundwater management.

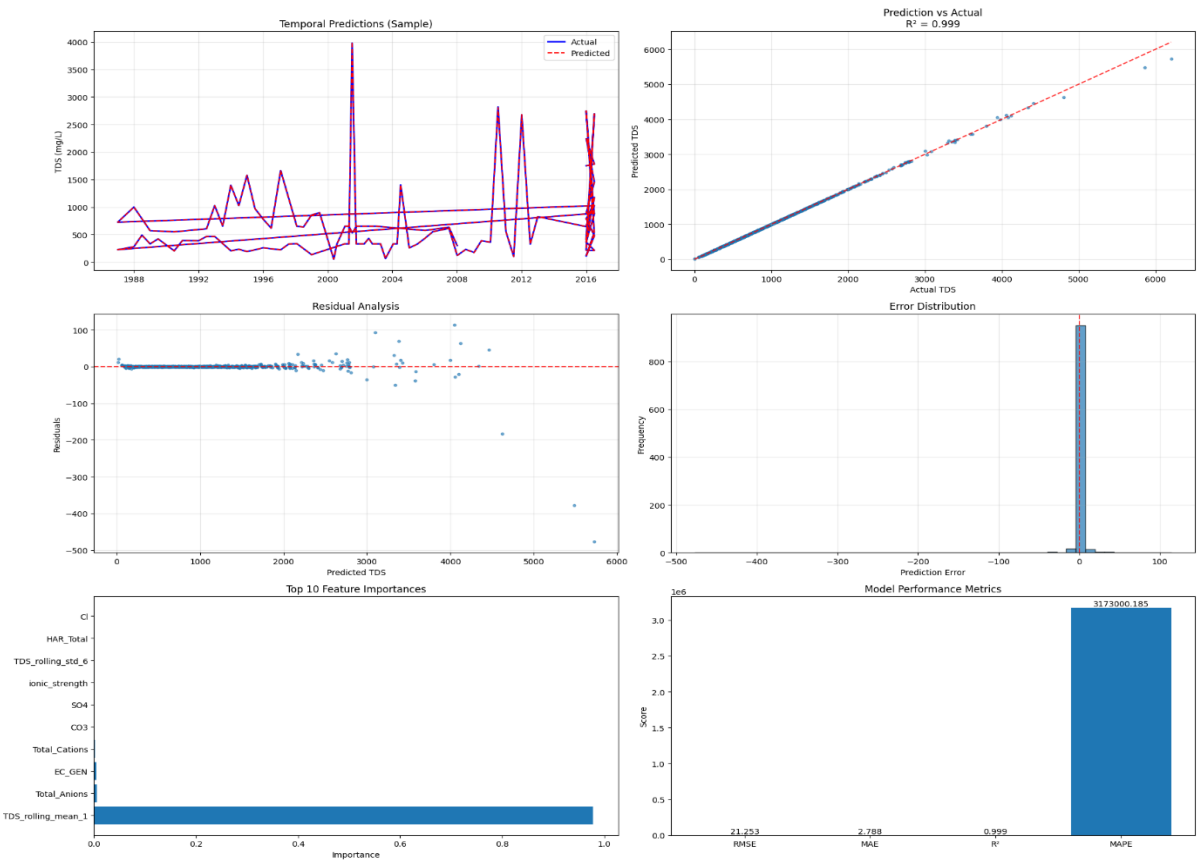

Figure 6.3 Performance Metrics

6.3 VISUALIZATION OF PREDICTED VS ACTUAL DATA

To visually validate the performance of the models, graphical comparisons between observed and predicted groundwater levels were generated. These visualizations provide intuitive confirmation of the model's ability to replicate seasonal fluctuations, recharge cycles, and depletion trends.

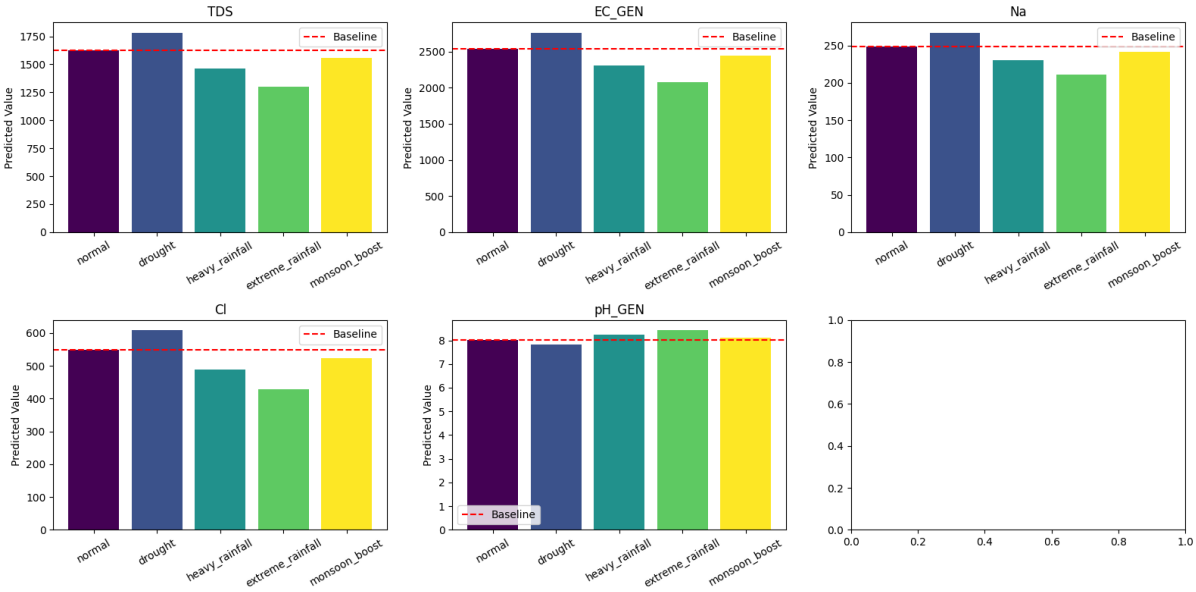

Figure 6.4 Parameter Specific Score Analysis

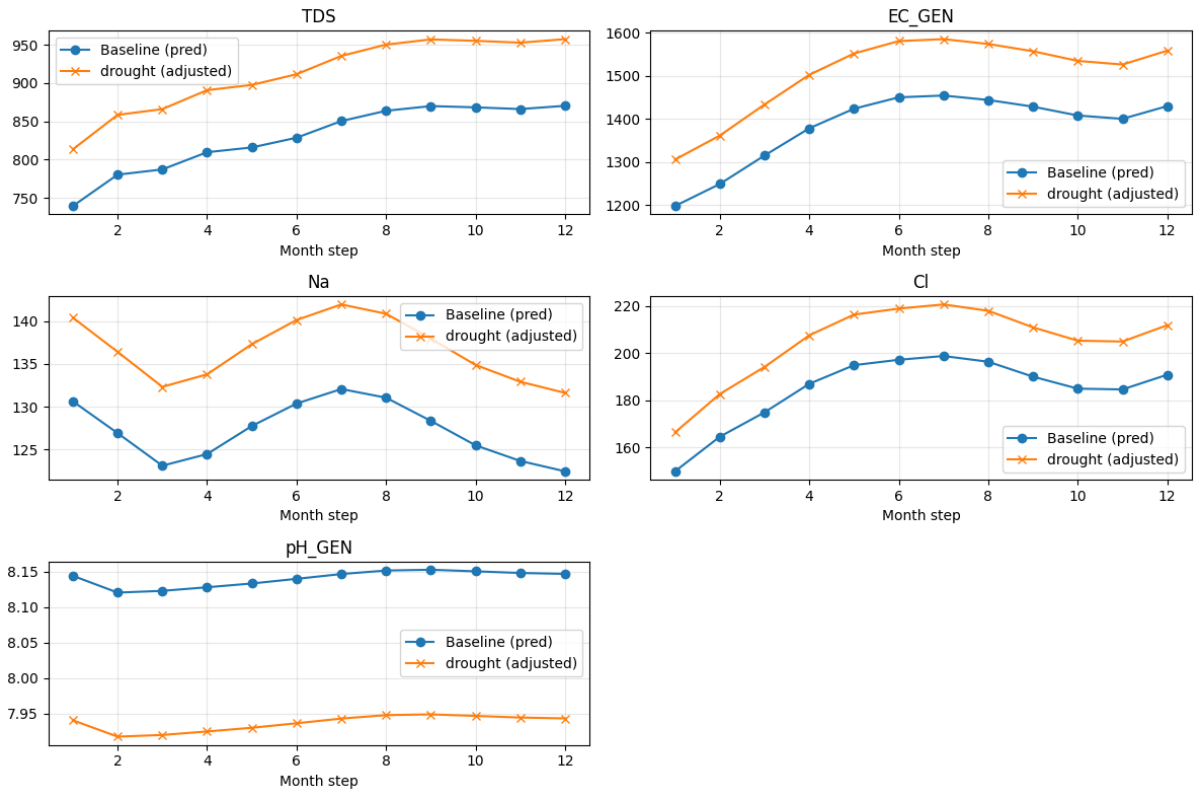

Figure 6.5 Scenario based Prediction 1

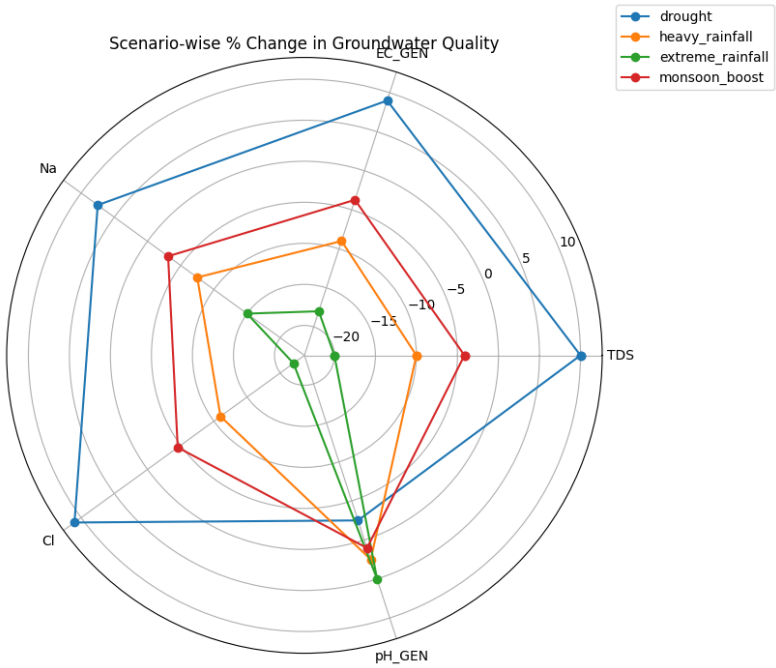

Figure 6.6 Parameter Specific % Change Analysis

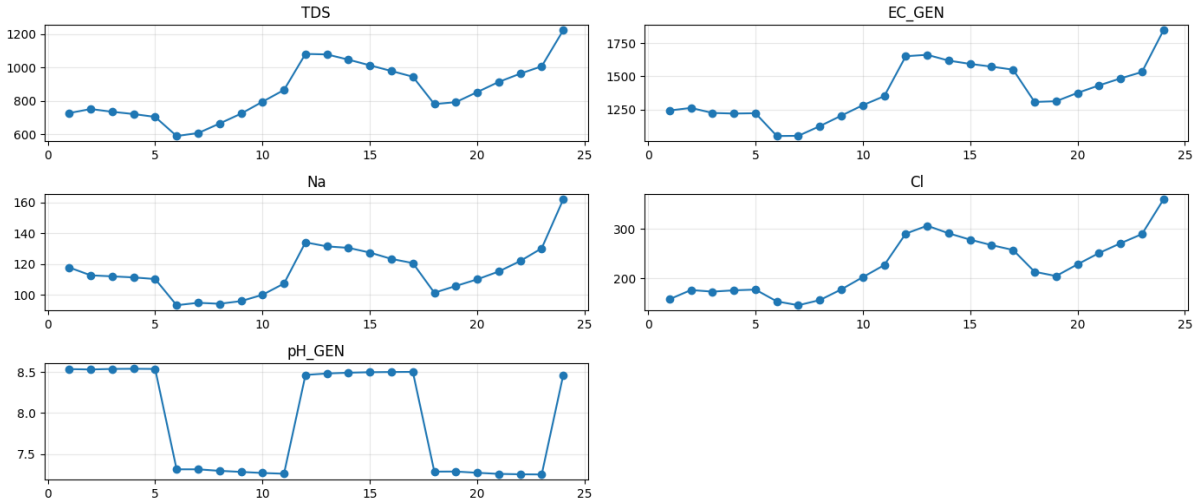

Figure 6.7 Scenario based Prediction 2

CHAPTER 7
CONCLUSION

The research presented in this study focused on the assessment and forecasting of groundwater resources in the context of climate change, with special reference to the Tiruppur District of Tamil Nadu. The increasing demand for groundwater, coupled with declining rainfall and industrialization, has made sustainable management of aquifers a major environmental and socio-economic challenge.

To address these issues, this work proposed a hybrid AI-driven forecasting framework integrating signal decomposition, deep learning, and adaptive optimization techniques. The framework combined the strengths of ICEEMDAN-VMD-SMA-LSTM, CNN-LSTM, and an Adaptive Weighting Model (AWM) to enhance prediction accuracy and stability under dynamic hydro-climatic conditions.

The proposed model demonstrated superior performance over conventional statistical and standalone deep learning approaches. The ICEEMDAN-VMD-SMA-LSTM model effectively handled non-stationary groundwater signals, while the CNN-LSTM captured spatial rainfall–groundwater relationships. The AWM ensemble outperformed all individual models, achieving the highest accuracy ($R^2 = 0.948$, NSE = 0.938) with the lowest RMSE (0.429 m).

Furthermore, visual analyses confirmed that the model closely followed observed temporal trends and spatial variations, validating its potential for practical application in real-time groundwater monitoring and climate-resilient water management.

By integrating climate projections (CMIP6 SSP2-4.5 and SSP5-8.5), the study also highlighted potential future groundwater declines under high-emission scenarios, emphasizing the urgent need for adaptive resource management in the Tiruppur region.

In summary, the developed hybrid forecasting system provides a scientifically robust, data-driven, and scalable solution for sustainable groundwater management and can be extended to other water-stressed regions across India.

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